



Maintenance Schedule

Diesel Engine

EPA/CARB

12V4000T94

12V4000T94L

Application group 4D

MSE50080/04E



A Rolls-Royce
solution

Engine model	kW/cyl.	Application group
12V4000T94	140.0 kW/cyl.	4D, Frac operation (continuous operation at low load)
12V4000T94L	155.4 kW/cyl.	4D, Frac operation (continuous operation at low load)

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Definition of Rolls-Royce Solutions

Rolls-Royce Solutions refers to Rolls-Royce Power Systems AG and its subsidiaries Rolls-Royce Solutions GmbH and Rolls-Royce Solutions America Inc.

Maintenance concept

The maintenance schedule is applicable to products certified in accordance with United States Environmental Protection Agency (U.S. EPA) and California Air Resources Board (CARB) emission regulations.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Non-observance of the maintenance instructions can result in violation of the applicable U.S. EPA and CARB emission regulations. Modification or removal of mechanical or electronic components, or the installation of additional components, as well as the execution of calibration processes which may affect the emissions characteristics of the product are prohibited by applicable U.S. EPA and/or CARB emission regulations.

Although Rolls-Royce Solutions' warranty obligations are not dependent upon the use of a specific brand of spare parts, it is recommended that emissions-relevant components be serviced, replaced or repaired using only Rolls-Royce Solutions approved components or their equivalent.

Emissions-relevant components include, but are not limited to, control units, data records, sensors, injectors, exhaust turbochargers, exhaust flaps and all components of the exhaust gas aftertreatment systems.

Use of spare parts that are not of equivalent quality to Rolls-Royce Solutions approved components may impair the effectiveness of the product's emission control systems. The maintenance, replacement, or repair of the emission control devices and systems may be performed by any repair establishment or individual of the owner's choosing.

Except for those maintenance steps designated in this Maintenance Schedule as critical emission-related maintenance, an owner or operator of the product is not required to perform the maintenance steps described in this Maintenance Schedule in order to keep the emission-related warranty valid. Rolls-Royce Solutions will not deny an emission-related warranty claim solely on the basis of an owner's or operator's failure to perform recommended additional maintenance or noncritical emission-related maintenance. However, Rolls-Royce Solutions may deny an emission-related warranty claim for the owner's or operator's failure to perform maintenance steps designated as critical emission-related maintenance or if the warranty claim arises out of failures in emission-related components that have been caused by the owner's or operator's improper maintenance or use, by accidents for which Rolls-Royce Solutions has no responsibility, or by acts of God.

If listed in this maintenance schedule, also partially, the following maintenance tasks are emission critical: WM00090, WM00150 and WM00151.

The maintenance system for products from Rolls-Royce Solutions is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals, as well as the required scope of maintenance tasks, are based on operational experience.

Adherence to the specified maintenance intervals is essential to maintain product safety.

Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

The intervals according to which the maintenance tasks must be carried out are specified in terms of operating hours and time limits. The specific maintenance task is triggered when either the operating hours or time limit is first reached.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by a Rolls-Royce Solutions' Service Partner. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Special conditions

Note

The engines are certified for operation with approved fuels as listed in the Fluids and Lubricants Specifications.

The maintenance intervals for the components specified in the maintenance schedule refer to engine operation with diesel fuel with a sulfur content < 15 ppm, e.g. EN 590.

If fuels with a sulfur content > 15 ppm are used, the maintenance intervals specified in the maintenance schedule for the affected components could be shorter.

Engines with exhaust gas recirculation and/or exhaust gas aftertreatment must not be operated with excessive sulfur content in the fuel. The limit values in the Fluids and Lubricants Specifications apply.

Table 2: Special conditions

1.3 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.4 Allocation matrix application group 4D

12V4000T94/T94L							
Load profile		Load factor %	Load indicator %	0 - 140 kW/cyl.	141 - 156 kW/cyl.	-	-
Load %	Time %						
110 ₍₁₀₀₋₁₁₀₎	—	0 - 52	≤ 12	15,000	9,000	—	—
100 ₍₉₀₋₁₀₀₎	5						
90 ₍₈₀₋₉₀₎	5						
80 ₍₆₅₋₈₀₎	40						
65 ₍₅₀₋₆₅₎	—						
50 ₍₃₀₋₅₀₎	—						
30 ₍₅₋₃₀₎	30						
Idle ₍₀₋₅₎	20						

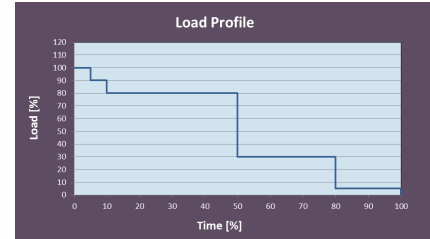


Table 3: Allocation matrix application group 4D

1.5 Noncyclical maintenance tasks

No.	Qualification	Task	Maintenance tasks	Reference task	Offset	Value	Unit	Option
0002	QL 1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		0	1000	HRS	
Reference task: Name of the maintenance task the offset refers to Offset: Interval of a maintenance task which depends on another interval Value: Fixed value for binding execution of a maintenance task								

1.6 Maintenance Schedule Matrix 15000 HRS

500 to 15000 Operating hours

Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00112	1 DAY																														
WM00233	1 DAY																														
WM00285	1 DAY																														
WM00286	1 DAY																														
WM00287	1 DAY																														
WM00288	1 DAY																														
WM00289	1 DAY																														
WM00290	1 DAY																														
WM00291	1 DAY																														
WM00174	1 WK																														
WM00881	1 MON																														
WM00126	1 YR																														
WM00128	1 YR																														
WM00067	2 YR																														
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00084	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00085	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00086	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00170	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00050	1 YR						X					X					X						X								X
WM00136	1 YR						X					X					X					X									X
WM00076	3 YR						X					X					X					X				X					X
WM00664	3 YR						X					X					X					X				X					X
WM00094	18 YR		X				X					X					X					X				X					X
WM00150	6 YR										X										X										X
WM00151	6 YR										X										X										X
WM00090	18 YR										X										X										X
WM00098	18 YR										X										X										X
WM00110	18 YR										X										X										X
WM00154	18 YR										X										X										X
WM00106	3 YR															X															X
WM00008	6 YR															X															X
WM00054	18 YR															X															X
WM00056	18 YR															X															X
WM00062	18 YR															X															X
WM00093	18 YR															X															X
WM00096	18 YR															X															X
WM00164	18 YR															X															X
WM00921	18 YR															X															X
WM00970	18 YR															X															X

DAY = Days
 WK = Weeks
 MON = Months
 YR = Years
 HRS = Hours

TIM-ID: 0000000000 - 000

Task	Limit	Operating hours [HRS]																														
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000	
WM00971	18 YR															X																X
WM00973	18 YR															X																X
WM00974	18 YR															X																X
WM00981	18 YR															X																X
WM00986	18 YR															X																X
WM00006	6 YR																															X
WM00001	18 YR																															X
WM00002	18 YR																															X
WM00011	18 YR																															X
WM00070	18 YR																															X
WM00072	18 YR																															X
WM00080	18 YR																															X
WM00087	18 YR																															X
WM00088	18 YR																															X
WM00091	18 YR																															X
WM00092	18 YR																															X
WM00095	18 YR																															X
WM00099	18 YR																															X
WM00100	18 YR																															X
WM00103	18 YR																															X
WM00104	18 YR																															X
WM00108	18 YR																															X
WM00111	18 YR																															X
WM00127	18 YR																															X
WM00137	18 YR																															X
WM00160	18 YR																															X
WM00283	18 YR																															X
WM00353	18 YR																															X
WM00439	18 YR																															X
WM00925	18 YR																															X
WM00945	18 YR																															X
WM00976	18 YR																															X
WM00980	18 YR																															X

DAY = Days
WK = Weeks
MON = Months
YR = Years
HRS = Hours

1.7 Cyclical maintenance tasks 15000 HRS

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL 1	-	1 DAY	PLATES AND MISCELLANEOUS MATERIALS	Check emissions label for presence, legibility and contents		WM00112
QL 1	-	1 DAY	EXHAUST SILENCER	Check drain of silencer	X	WM00233
QL 1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check engine oil level		WM00285
QL 1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Visually inspect for leaks and general condition of engine		WM00286
QL 1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check maintenance indicator of air filter	X	WM00287
QL 1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check relief bore of coolant pump		WM00288
QL 1	-	1 DAY	ENGINE OPERATIONAL MONITORING	When engine is running, check for abnormal running noises, exhaust gas color and vibrations		WM00289
QL 1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Drain water and contaminants from fuel prefilter	X	WM00290
QL 1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check position of contamination indicator of fuel prefilter	X	WM00291
QL 1	-	1 WK	EMERGENCY AIR SHUT-OFF FLAPS	Check operation of emergency air shut-off flaps	X	WM00174
QL 1	-	1 MON	BENDING / TORS. RESILIENT COUPLING AND STEEL-SPRING COUPLING	Check coupling (visual inspection), replace as necessary	X	WM00881
QL 1	-	1 YR	ENGINE MOUNTING	Measure height of rubber element		WM00126
QL 1	-	1 YR	ENGINE MOUNTING	Measure buffer play, adjust if necessary		WM00128
QL 1	-	2 YR	LUBE OIL SYSTEM	Replace oil filter at every engine oil change, or when the time limit (years) is reached, at the latest		WM00067
QL 1	500	2 YR	LUBE OIL SYSTEM	Automatic centrifugal oil filter, check layer thickness of oil residue, clean, and replace paper element at every engine oil change at the latest	X	WM00063
QL 1	1000	2 YR	FUEL SYSTEM (LOW PRESSURE)	Fuel prefilter, replace filter	X	WM00084
QL 1	1000	2 YR	FUEL SYSTEM (LOW PRESSURE)	Additional filter, replace easy-change fuel filter	X	WM00085
QL 1	1000	2 YR	FUEL SYSTEM (LOW PRESSURE)	Replace easy-change fuel filter		WM00086
QL 1	1000	2 YR	AUXILIARY POWER TAKE-OFF	Check belt condition, replace if necessary. Adjust tension.	X	WM00170
QL 1	3000	1 YR	ENGINE MOUNTING	Check general condition of rubber mounts (visual inspection)		WM00050
QL 1	3000	1 YR	ENGINE MOUNTING	Check securing screws for secure seating		WM00136
QL 1	3000	3 YR	AIR SUPPLY SYSTEM	Replace air filter	X	WM00076
QL 1	3000	3 YR	COOLANT SYSTEM	Replace valve cover	X	WM00664
QL 1	3000	18 YR	VALVE GEAR	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		WM00094
QL 1	5000	6 YR	SENSOR TECHNOLOGY AND WIRING HARNESS	Replace NOx sensor(s)		WM00150
QL 1	5000	6 YR	SENSOR TECHNOLOGY AND WIRING HARNESS	Replace Lambda sensor		WM00151
QL 1	5000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace fuel injection valve/injector		WM00090

DAY = Days
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MON = Months
YR = Years
HRS = Hours

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Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
QL1	5000	18 YR	CYLINDER HEAD	Measure valve protrusion		WM00098
QL1	5000	18 YR	RUNNING GEAR	Inspect cylinder liner with endoscope		WM00110
QL1	5000	18 YR	GOVERNOR	Reset parameters for drift correction (CDC) and enter injector coding (IIG)		WM00154
QL1	7500	3 YR	CRANKCASE VENTILATION	Replace oil separator (impactor)		WM00106
QL3	7500	6 YR	RUBBER SLEEVE	Replace rubber sleeve		WM00008
QL3	7500	18 YR	COOLANT SYSTEM	Check LT thermostat housing and replace thermal actuator		WM00054
QL3	7500	18 YR	COOLANT SYSTEM	Check HT thermostat housing and replace thermal actuator		WM00056
QL3	7500	18 YR	LUBE OIL SYSTEM	Check centrifugal oil filter for wear	X	WM00062
QL3	7500	18 YR	VALVE GEAR	Check rocker arm and valve bridge for wear. Visually inspect swing follower with endoscope through pushrod bore.		WM00093
QL3	7500	18 YR	VALVE GEAR	Visually inspect camshaft running surfaces with endoscope through pushrod bore		WM00096
QL3	7500	18 YR	AIR PIPING TO CYLINDERS	Clean air piping		WM00164
QL3	7500	18 YR	COOLANT SYSTEM	Replace sealing material for installation of coolant pump(s) and thermostat(s) (only LT and HT)		WM00921
QL3	7500	18 YR	EXHAUST TURBOCHARGER	Overhaul or replace exhaust turbocharger		WM00970
QL3	7500	18 YR	STARTING SYSTEM	Overhaul or replace starter		WM00971
QL3	7500	18 YR	COOLANT SYSTEM	Overhaul or replace coolant pump in LT circuit		WM00973
QL3	7500	18 YR	COOLANT SYSTEM	Overhaul or replace coolant pump in HT circuit		WM00974
QL3	7500	18 YR	LUBE OIL SYSTEM	Clean oil cooler and check for leaks or replace		WM00981
QL3	7500	18 YR	AIR PIPING TO CYLINDERS	Clean intercooler and check for leaks or replace		WM00986
QL3	15000	6 YR	HOSE	Replace hose line		WM00006
QL4	15000	18 YR	ENGINE, GENERAL	Replace elastomeric members and gaskets		WM00001
QL4	15000	18 YR	ENGINE, GENERAL	Disassemble engine completely. Check engine components according to assembly instructions, replace or repair as necessary		WM00002
QL4	15000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace HP fuel sensor		WM00011
QL4	15000	18 YR	LUBE OIL SYSTEM	Check pump, replace if necessary		WM00070
QL4	15000	18 YR	EXHAUST SYSTEM	Replace exhaust pipe bellows		WM00072
QL4	15000	18 YR	EXHAUST TURBOCHARGER	Replace aluminum compressor wheel of secondary exhaust turbocharger		WM00080
QL4	15000	18 YR	FUEL SYSTEM (LOW PRESSURE)	Replace pump		WM00087
QL4	15000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace pressure relief valve		WM00088
QL4	15000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace HP pump		WM00091
QL4	15000	18 YR	VALVE GEAR	Replace swing follower and swing follower shaft		WM00092
QL4	15000	18 YR	VALVE GEAR	Replace thrust bearing flange		WM00095
QL4	15000	18 YR	RUNNING GEAR	Replace conrod bearings		WM00099
QL4	15000	18 YR	RUNNING GEAR	Replace rings		WM00100
QL4	15000	18 YR	RUNNING GEAR	Replace crankshaft bearings		WM00103
QL4	15000	18 YR	GEAR TRAIN	Check tooth flanks in gear train for wear (visual inspection), replace bushing		WM00104
QL4	15000	18 YR	RUNNING GEAR	Replace cylinder liners		WM00108
QL4	15000	18 YR	CAMSHAFT	Replace camshaft bearings		WM00111
QL4	15000	18 YR	ENGINE MOUNTING	Replace rubber element of rubber mounts		WM00127
QL4	15000	18 YR	SENSOR TECHNOLOGY AND WIRING HARNESS	Replace wiring harness		WM00137

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HRS = Hours

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
QL4	15000	18 YR	EXHAUST RETURN	Replace flaps		WM00160
QL4	15000	18 YR	AUXILIARY POWER TAKE-OFF	Replace auxiliary power take-off	X	WM00283
QL4	15000	18 YR	RUNNING GEAR	Replace conrod		WM00353
QL4	15000	18 YR	RUNNING GEAR	Replace vibration damper		WM00439
QL4	15000	18 YR	BENDING / TORSIONALLY RESILIENT COUPLING	Replace coupling or coupling element(s)	X	WM00925
QL4	15000	18 YR	EXHAUST RETURN	Replace EGR cooler		WM00945
QL4	15000	18 YR	GENERATOR	Overhaul or replace battery-charging generator	X	WM00976
QL4	15000	18 YR	CYLINDER HEAD	Overhaul or replace cylinder head		WM00980
DAY = Days WK = Weeks MON = Months YR = Years HRS = Hours						

1.8 Maintenance Schedule Matrix 9000 HRS

500 to 9000 Operating hours

Task	Limit	Operating hours [HRS]																											
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000										
WM00112	1 DAY																												
WM00233	1 DAY																												
WM00285	1 DAY																												
WM00286	1 DAY																												
WM00287	1 DAY																												
WM00288	1 DAY																												
WM00289	1 DAY																												
WM00290	1 DAY																												
WM00291	1 DAY																												
WM00174	1 WK																												
WM00881	1 MON																												
WM00126	1 YR																												
WM00128	1 YR																												
WM00067	2 YR																												
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X								
WM00084	2 YR		X		X		X		X		X		X		X		X		X		X								
WM00085	2 YR		X		X		X		X		X		X		X		X		X		X								
WM00086	2 YR		X		X		X		X		X		X		X		X		X		X								
WM00170	2 YR		X		X		X		X		X		X		X		X		X		X								
WM00050	1 YR						X						X							X									
WM00136	1 YR						X						X							X									
WM00076	3 YR						X						X							X									
WM00664	3 YR						X						X							X									
WM00094	18 YR		X				X						X							X									
WM00150	6 YR									X										X									
WM00151	6 YR									X										X									
WM00090	18 YR									X										X									
WM00098	18 YR									X										X									
WM00110	18 YR									X										X									
WM00154	18 YR									X										X									
WM00106	3 YR																			X									
WM00008	6 YR									X										X									
WM00054	18 YR									X										X									
WM00056	18 YR									X										X									
WM00062	18 YR									X										X									
WM00093	18 YR									X										X									
WM00096	18 YR									X										X									
WM00164	18 YR									X										X									
WM00921	18 YR									X										X									
WM00970	18 YR									X										X									
WM00971	18 YR									X										X									

DAY = Days
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 MON = Months
 YR = Years
 HRS = Hours

Task	Limit	Operating hours [HRS]																	
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000
WM00973	18 YR									X									X
WM00974	18 YR									X									X
WM00981	18 YR									X									X
WM00986	18 YR									X									X
WM00006	6 YR																		X
WM00001	18 YR																		X
WM00002	18 YR																		X
WM00011	18 YR																		X
WM00070	18 YR																		X
WM00072	18 YR																		X
WM00080	18 YR																		X
WM00087	18 YR																		X
WM00088	18 YR																		X
WM00091	18 YR																		X
WM00092	18 YR																		X
WM00095	18 YR																		X
WM00099	18 YR																		X
WM00100	18 YR																		X
WM00103	18 YR																		X
WM00104	18 YR																		X
WM00108	18 YR																		X
WM00111	18 YR																		X
WM00127	18 YR																		X
WM00137	18 YR																		X
WM00160	18 YR																		X
WM00283	18 YR																		X
WM00353	18 YR																		X
WM00439	18 YR																		X
WM00925	18 YR																		X
WM00945	18 YR																		X
WM00976	18 YR																		X
WM00980	18 YR																		X

DAY = Days

WK = Weeks

MON = Months

YR = Years

HRS = Hours

1.9 Cyclical maintenance tasks 9000 HRS

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	1 DAY	PLATES AND MISCELLANEOUS MATERIALS	Check emissions label for presence, legibility and contents		WM00112
QL1	-	1 DAY	EXHAUST SILENCER	Check drain of silencer	X	WM00233
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check engine oil level		WM00285
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Visually inspect for leaks and general condition of engine		WM00286
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check maintenance indicator of air filter	X	WM00287
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check relief bore of coolant pump		WM00288
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	When engine is running, check for abnormal running noises, exhaust gas color and vibrations		WM00289
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Drain water and contaminants from fuel prefilter	X	WM00290
QL1	-	1 DAY	ENGINE OPERATIONAL MONITORING	Check position of contamination indicator of fuel prefilter	X	WM00291
QL1	-	1 WK	EMERGENCY AIR SHUT-OFF FLAPS	Check operation of emergency air shut-off flaps	X	WM00174
QL1	-	1 MON	BENDING / TORS. RESILIENT COUPLING AND STEEL-SPRING COUPLING	Check coupling (visual inspection), replace as necessary	X	WM00881
QL1	-	1 YR	ENGINE MOUNTING	Measure height of rubber element		WM00126
QL1	-	1 YR	ENGINE MOUNTING	Measure buffer play, adjust if necessary		WM00128
QL1	-	2 YR	LUBE OIL SYSTEM	Replace oil filter at every engine oil change, or when the time limit (years) is reached, at the latest		WM00067
QL1	500	2 YR	LUBE OIL SYSTEM	Automatic centrifugal oil filter, check layer thickness of oil residue, clean, and replace paper element at every engine oil change at the latest	X	WM00063
QL1	1000	2 YR	FUEL SYSTEM (LOW PRESSURE)	Fuel prefilter, replace filter	X	WM00084
QL1	1000	2 YR	FUEL SYSTEM (LOW PRESSURE)	Additional filter, replace easy-change fuel filter	X	WM00085
QL1	1000	2 YR	FUEL SYSTEM (LOW PRESSURE)	Replace easy-change fuel filter		WM00086
QL1	1000	2 YR	AUXILIARY POWER TAKE-OFF	Check belt condition, replace if necessary. Adjust tension.	X	WM00170
QL1	3000	1 YR	ENGINE MOUNTING	Check general condition of rubber mounts (visual inspection)		WM00050
QL1	3000	1 YR	ENGINE MOUNTING	Check securing screws for secure seating		WM00136
QL1	3000	3 YR	AIR SUPPLY SYSTEM	Replace air filter	X	WM00076
QL1	3000	3 YR	COOLANT SYSTEM	Replace valve cover	X	WM00664
QL1	3000	18 YR	VALVE GEAR	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		WM00094
QL1	4500	6 YR	SENSOR TECHNOLOGY AND WIRING HARNESS	Replace NOx sensor(s)		WM00150
QL1	4500	6 YR	SENSOR TECHNOLOGY AND WIRING HARNESS	Replace Lambda sensor		WM00151
QL1	4500	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace fuel injection valve/injector		WM00090

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 HRS = Hours

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
QL1	4500	18 YR	CYLINDER HEAD	Measure valve protrusion		WM00098
QL1	4500	18 YR	RUNNING GEAR	Inspect cylinder liner with endoscope		WM00110
QL1	4500	18 YR	GOVERNOR	Reset parameters for drift correction (CDC) and enter injector coding (IIG)		WM00154
QL1	9000	3 YR	CRANKCASE VENTILATION	Replace oil separator (impactor)		WM00106
QL3	4500	6 YR	RUBBER SLEEVE	Replace rubber sleeve		WM00008
QL3	4500	18 YR	COOLANT SYSTEM	Check LT thermostat housing and replace thermal actuator		WM00054
QL3	4500	18 YR	COOLANT SYSTEM	Check HT thermostat housing and replace thermal actuator		WM00056
QL3	4500	18 YR	LUBE OIL SYSTEM	Check centrifugal oil filter for wear	X	WM00062
QL3	4500	18 YR	VALVE GEAR	Check rocker arm and valve bridge for wear. Visually inspect swing follower with endoscope through pushrod bore.		WM00093
QL3	4500	18 YR	VALVE GEAR	Visually inspect camshaft running surfaces with endoscope through pushrod bore		WM00096
QL3	4500	18 YR	AIR PIPING TO CYLINDERS	Clean air piping		WM00164
QL3	4500	18 YR	COOLANT SYSTEM	Replace sealing material for installation of coolant pump(s) and thermostat(s) (only LT and HT)		WM00921
QL3	4500	18 YR	EXHAUST TURBOCHARGER	Overhaul or replace exhaust turbocharger		WM00970
QL3	4500	18 YR	STARTING SYSTEM	Overhaul or replace starter		WM00971
QL3	4500	18 YR	COOLANT SYSTEM	Overhaul or replace coolant pump in LT circuit		WM00973
QL3	4500	18 YR	COOLANT SYSTEM	Overhaul or replace coolant pump in HT circuit		WM00974
QL3	4500	18 YR	LUBE OIL SYSTEM	Clean oil cooler and check for leaks or replace		WM00981
QL3	4500	18 YR	AIR PIPING TO CYLINDERS	Clean intercooler and check for leaks or replace		WM00986
QL3	9000	6 YR	HOSE	Replace hose line		WM00006
QL4	9000	18 YR	ENGINE, GENERAL	Replace elastomeric members and gaskets		WM00001
QL4	9000	18 YR	ENGINE, GENERAL	Disassemble engine completely. Check engine components according to assembly instructions, replace or repair as necessary		WM00002
QL4	9000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace HP fuel sensor		WM00011
QL4	9000	18 YR	LUBE OIL SYSTEM	Check pump, replace if necessary		WM00070
QL4	9000	18 YR	EXHAUST SYSTEM	Replace exhaust pipe bellows		WM00072
QL4	9000	18 YR	EXHAUST TURBOCHARGER	Replace aluminum compressor wheel of secondary exhaust turbocharger		WM00080
QL4	9000	18 YR	FUEL SYSTEM (LOW PRESSURE)	Replace pump		WM00087
QL4	9000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace pressure relief valve		WM00088
QL4	9000	18 YR	FUEL SYSTEM (HIGH PRESSURE)	Replace HP pump		WM00091
QL4	9000	18 YR	VALVE GEAR	Replace swing follower and swing follower shaft		WM00092
QL4	9000	18 YR	VALVE GEAR	Replace thrust bearing flange		WM00095
QL4	9000	18 YR	RUNNING GEAR	Replace conrod bearings		WM00099
QL4	9000	18 YR	RUNNING GEAR	Replace rings		WM00100
QL4	9000	18 YR	RUNNING GEAR	Replace crankshaft bearings		WM00103
QL4	9000	18 YR	GEAR TRAIN	Check tooth flanks in gear train for wear (visual inspection), replace bushing		WM00104
QL4	9000	18 YR	RUNNING GEAR	Replace cylinder liners		WM00108
QL4	9000	18 YR	CAMSHAFT	Replace camshaft bearings		WM00111
QL4	9000	18 YR	ENGINE MOUNTING	Replace rubber element of rubber mounts		WM00127
QL4	9000	18 YR	SENSOR TECHNOLOGY AND WIRING HARNESS	Replace wiring harness		WM00137

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TIM-ID: 0000000000 - 000

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
QL4	9000	18 YR	EXHAUST RETURN	Replace flaps		WM00160
QL4	9000	18 YR	AUXILIARY POWER TAKE-OFF	Replace auxiliary power take-off	X	WM00283
QL4	9000	18 YR	RUNNING GEAR	Replace conrod		WM00353
QL4	9000	18 YR	RUNNING GEAR	Replace vibration damper		WM00439
QL4	9000	18 YR	BENDING / TORSIONALLY RESILIENT COUPLING	Replace coupling or coupling element(s)	X	WM00925
QL4	9000	18 YR	EXHAUST RETURN	Replace EGR cooler		WM00945
QL4	9000	18 YR	GENERATOR	Overhaul or replace battery-charging generator	X	WM00976
QL4	9000	18 YR	CYLINDER HEAD	Overhaul or replace cylinder head		WM00980
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